

STAT 516: Homework 7

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(10 points)** Let X be a continuous random variable with density function, for $b > 0$,

$$f(x) = \begin{cases} be^{-bx}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The MGF of X is denoted by $M_X(t)$. What is $M_X(-6b)$?

2. **(13 points)** Let X have the PMF

$$P_X(x) = \begin{cases} \left(\frac{8}{9}\right)^x \left(\frac{1}{9}\right), & x = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

What is the MGF of X ?

3. **(12 points)** The moment generating function for X is

$$M(t) = c + 0.4e^t + 0.3e^{2t} + 0.2e^{at},$$

where c is a constant. The mean of the variable is 2. Determine a .

4. **(13 points)** Let the random variable X have moment generating function

$$M(t) = e^{3t+t^2}.$$

What is $E[X^2]$?

5. **(13 points)** Let X be a random variable with moment generating function

$$M(t) = \left(\frac{2 + e^t}{3}\right)^9.$$

Calculate the variance of X .

6. **(13 points)** An actuary determines that the claim size for a certain class of accidents is a random variable X with moment generating function

$$M(t) = \frac{1}{(1 - 2500t)^4}.$$

Calculate the standard deviation of the claim size for this class of accidents.

7. **(13 points)** Two claimants place calls simultaneously to an insurer's claims call center. The times X and Y , in minutes, that elapse before the respective claimants get to speak with call center representatives are independently and identically distributed. The moment generating function of each random variable is

$$M(t) = \frac{1}{(1 - 1.5t)^2}, \quad t < \frac{2}{3}.$$

Calculate the standard deviation of $X + Y$.

8. **(13 points)** The moment generating function of X is

$$M(t) = \frac{1}{(1 - 0.1t)^3}.$$

Let $Y = 3X + 5$. Let $M_Y(t)$ be the moment generating function of Y . Determine $M_Y(0.5)$.